

Flutter Fundamentals

Short recap day #2

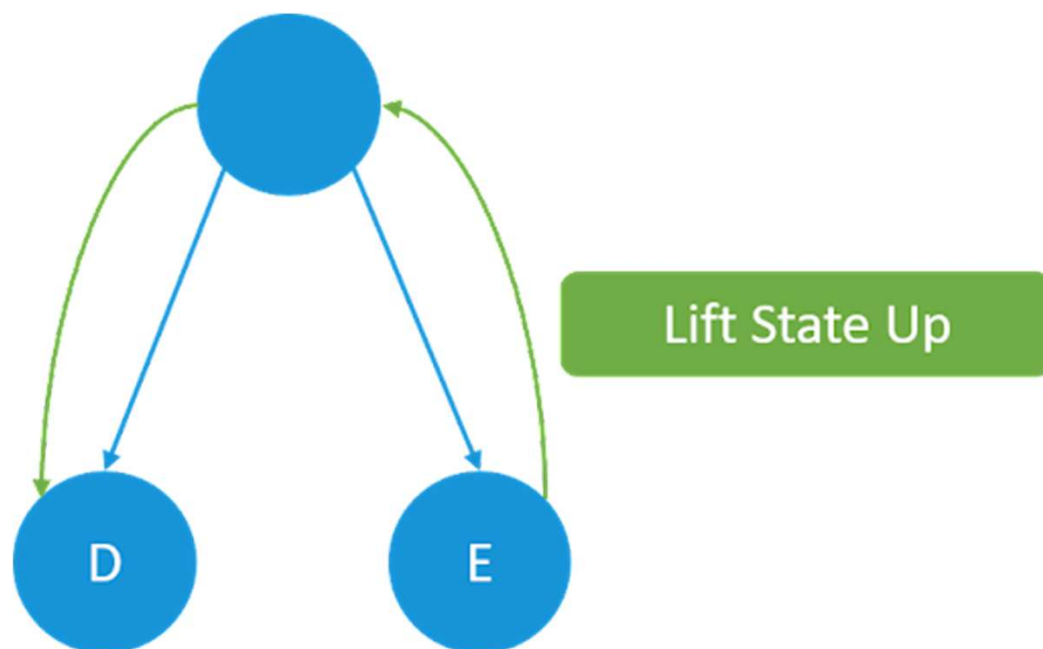


Peter Kassenaar –
info@kassenaar.com



Yesterday:

- **State management** in various ways
 - Stateful widgets – using models/custom classes
 - passing parameters, passing functions
 - Design principle: “Lifting state up”



two types of widgets

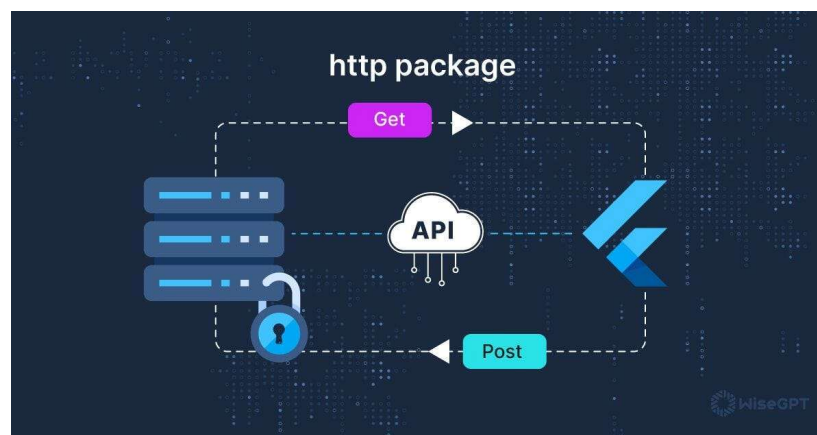


- Stateless Widgets, Stateful Widgets
 - Stateful Widgets are actually two classes:
- 1. Retain state between repaints, so widget's internal state survives hot reloads and rebuilds.
- 2. Optimize runtime performance, immutable widgets are fast to diff and rebuild;

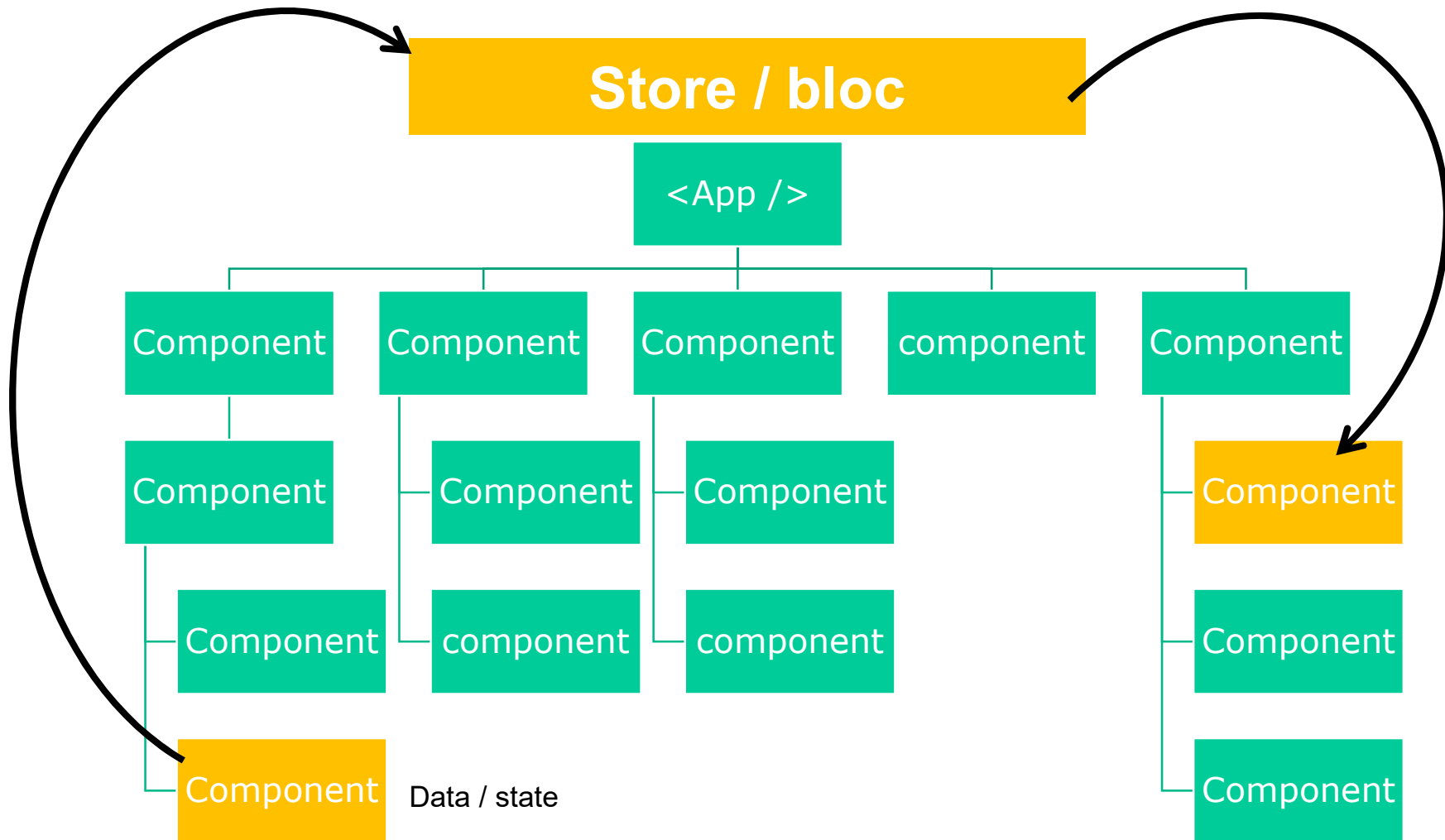


Also yesterday:

- Creating and using Custom Classes, Extracting Widgets
 - Passing parameters
 - Passing functions
- Communicating with external API's
 - http / other methods



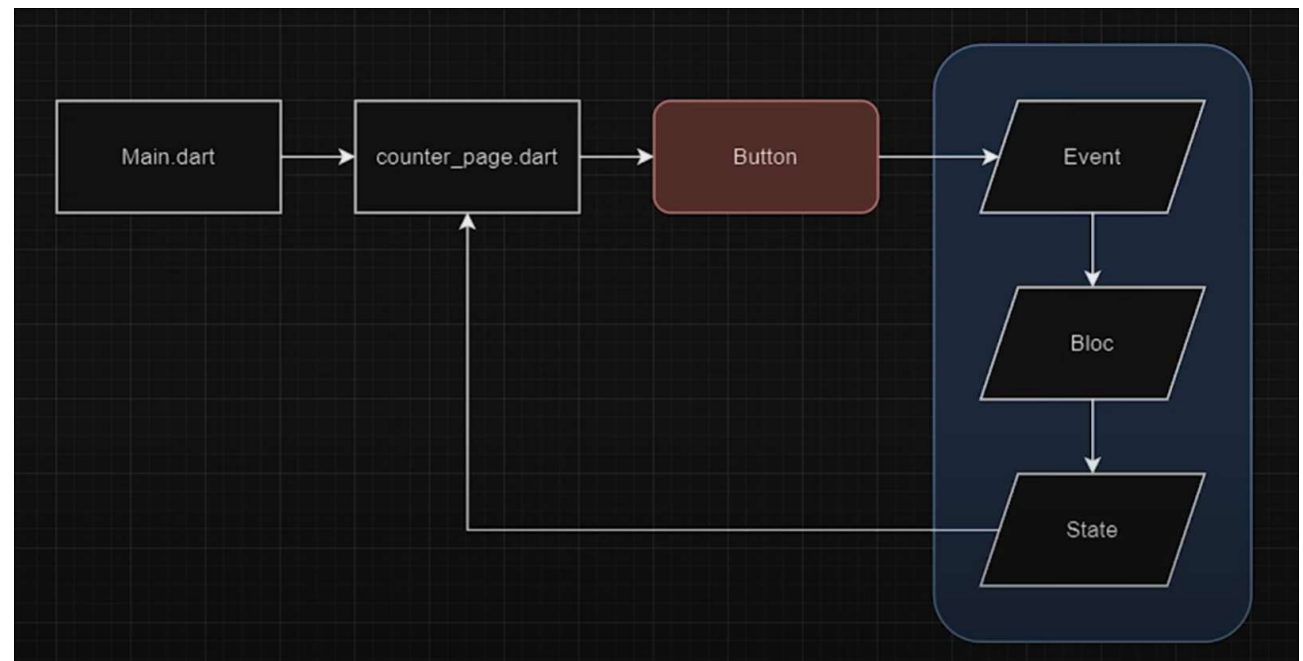
State management with a store / bloc



Also yesterday



- State management: share application state between unconnected Widgets/components
- Bloc pattern
 - State
 - Events
 - Bloc
 - Page





Bloc pattern

- Using `xxx_event.dart`
 - Define which events are available for page + bloc.
- Using `xxx_bloc.dart`
 - Listen to events, update state based on events
- Using `xxx_state.dart;`
 - Define which properties belong to the state
 - Updated from a bloc-page.
- Using `xxx_page.dart`
 - Read from the state + add events to the `xxxBloc()`.

Using Cubit:



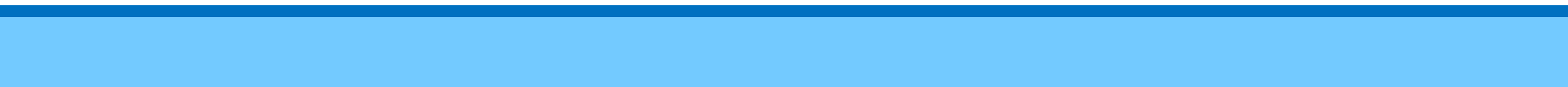
"Cubit is a lighter-weight abstraction than bloc, that doesn't rely on events"

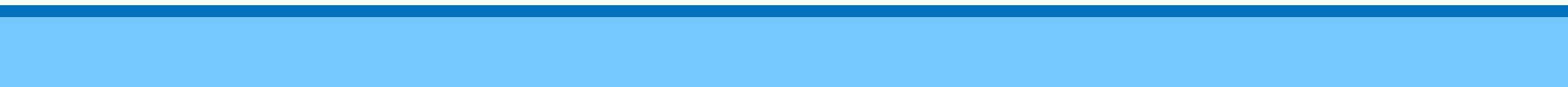
SO:

- `XXX_cubit.dart`
 - replacing both `XXX_bloc.dart` and `XXX_event.dart`
- `XXX_state.dart`



Questions?

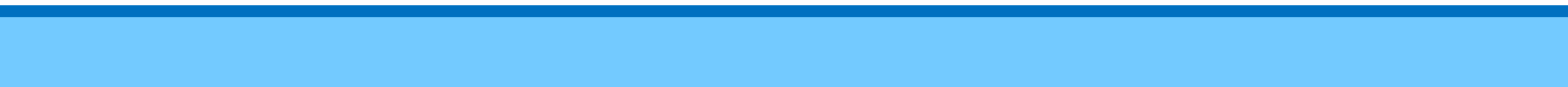




Today



- Using `TextFields`
- Routing / Navigation
- Complete applications using bloc/navigation/state
- Gestures
- gRPC - theory
- Deployment
- Evals & goodbye





More info

More Flutter info on the internet

Learn the widgets



Flutter > material > AboutDialog class

AboutDialog class

An about box. This is a dialog box with the application's icon, name, version number, and copyright, plus a button to show licenses for software used by the application.

To show an AboutDialog, use `showAboutDialog`.

The banner features the Flutter logo on the left, followed by the text "AboutDialog (Flutter Widget of the Week)". On the right, there are two buttons: "Later bekij..." and "Delen". The main text "Widget of the Week" is in a light blue font, and "AboutDialog" is in a large white font. A play button icon is centered over the text.

CLASSES

- AboutDialog
- AboutListTile
- AbsorbPointer
- Accumulator
- Action
- ActionChip
- ActionDispatcher
- ActionListener
- Actions
- ActivateAction
- ActivateIntent
- AlertDialog
- Align
- Alignment
- AlignmentDirectional
- AlignmentGeometry
- AlignmentGeometryTween
- AlignmentTween
- AlignTransition
- AlwaysScrollableScrollPh...

<https://api.flutter.dev/>

Stackoverflow tags



The screenshot displays the Stack Overflow interface for the 'flutter' tag. The left sidebar contains navigation links: Home, PUBLIC, Stack Overflow (selected), Tags, Users, FIND A JOB (Jobs, Companies), TEAMS, and What's this? (Free 30 Day Trial). The main content area is titled 'Questions tagged [flutter]' and includes an 'Ask Question' button. Below the title is a description of Flutter: 'Flutter is an open-source UI software development kit created by Google. It is used to develop applications for Android, iOS, Linux, Mac, Windows, Google Fuchsia and the web from a single codebase. Flutter apps are written in the Dart language.' There are buttons for 'Watch tag' and 'Ignore tag', and links for 'Learn more...', 'Improve tag info', 'Top users', and 'Synonyms'. A summary shows '73,753 questions' with filters for 'Newest', 'Active', 'Bountied 6', 'Unanswered', and 'More'. A section indicates '1 question with new activity'. Three questions are listed:

- How do i navigate from one activity to another in flutter**
0 votes, 0 answers, 3 views. Asked 1 min ago by Rishail Siddiqui (29 ● 6). Tags: android, flutter, dart.
- Issues getting multiple values from List of Maps**
0 votes, 0 answers, 4 views. Asked 4 mins ago by Michael Knight (15 ● 1 ● 6). Tags: json, flutter, sorting, dart, hashmap.
- GestureDetector OnTap Delay on Android Studio / Flutter / Nexus 6 API 30**
0 votes, 1 answer, 10 views. Asked 17 mins ago by Nurol Alacaatli (35 ● 4). Tags: android-studio, flutter, gesturedetector.

The right sidebar contains 'The Overflow Blog' with links to 'Fulfilling the promise of CI/CD' and 'Podcast 305: What does it mean to be a "senior" software engineer'. It also features 'Featured on Meta' (2020: a year in moderation, Should the [which] tag be burninated?), 'Hot Meta Posts' (63 Closing an old upvoted question as duplicate of new unvoted questions), 'Custom Filters' (Create a custom filter), 'Watched Tags' (angular, javascript, typescript), 'Ignored Tags' (Add an ignored tag), and 'Related Tags' (dart x 34083).

<https://stackoverflow.com/questions/tagged/flutter>

YouTube



The screenshot shows a YouTube playlist interface. On the left is a sidebar with navigation options: Home, Trending, Abonnementen, Bibliotheek, Geschiedenis, Je video's, Je films, Later bekijken, and Meer weergeven. Below this is a section for 'ABONNEMENTEN' (Subscriptions) listing channels like The Net Ninja, NOS Sport, Scott Hanselman, Global Cycling Net..., A&P-REACTS, and WatchMojo.com.

The main content area features a video player for 'Introducing Flutter' by Google Developers. Below the player, it says 'Flutter by Google' and provides statistics: '84 video's • 385.146 weergaven • Laatste geüpdatet op 12 nov. 2019'. It includes a 'Get started at <https://flutter.io> today.' link and a description of Flutter as Google's mobile UI framework.

On the right, a list of 7 videos in the playlist is shown:

1. **Introducing Flutter** (Google Developers, 1:53)
2. **Alibaba used Flutter to build 50+ million user Xianyu app (Flutter Developer Story)** (Google Developers, 2:51)
3. **Hamilton app built with Flutter and featured on iOS and Android (Flutter Developer Story)** (Google Developers, 2:37)
4. **Abbey Road Studios (Flutter Developer Story)** (Google Developers, 3:16)
5. **Flutter Live Keynote Recap** (Google Developers, 11:09)
6. **Flutter Release Preview 2** (Google Developers, 3:01)
7. **Building your first Flutter Widget** (Google Developers, 10:05)

<https://www.youtube.com/playlist?list=PLOU2XLYxmsIJ7dsVN4iRuA7BT8XHzGtCr>

Firestore



The screenshot shows the Firebase documentation website. The top navigation bar includes links for Products, Use Cases, Pricing, Docs (selected), Community, and Support. A search bar and language selector (English) are also present. The left sidebar lists various documentation categories, with 'Flutter' highlighted under the 'Guides' section. The main content area is titled 'Add Firebase to your Flutter app' and includes a video thumbnail for 'Getting started with Firebase on Flutter'. A table of contents on the right lists the steps: Create a Firebase project, Register your app with Firebase, Add a Firebase configuration file, and Add FlutterFire plugins.

Documentation

Overview Guides Reference Samples Libraries

Guides

- Get started with Firestore
 - Add Firestore to an app
 - Add Firestore to a game
 - Add Firestore to a server
 - Use Firestore with a framework
 - Flutter**
 - Set up projects programmatically
- Manage your Firestore projects
- Platforms and frameworks
- Prototype and test with Emulator Suite
- Analytics
- Extensions
- BUILD
- Authentication

Firestore is committed to advancing racial equity for Black communities. [See how.](#)

Firestore > Docs > Guides

Add Firestore to your Flutter app

Follow this guide to add Firestore products to a Flutter app.

Note: Firestore supports frameworks like Flutter on a best-effort basis. These integrations are not covered by Firestore Support and may not have full feature parity with the [official Firestore SDKs](#).

Prerequisites

- Install your preferred [editor or IDE](#).

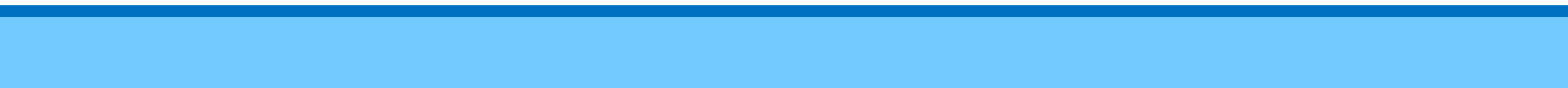
Table of contents

- [Prerequisites](#)
- Step 1: Create a Firestore project
- Step 2: Register your app with Firestore
- Step 3: Add a Firestore configuration file
- Step 4: Add FlutterFire plugins
- Try out an example app with Analytics
- Next steps

<https://firebase.google.com/docs/flutter/setup>



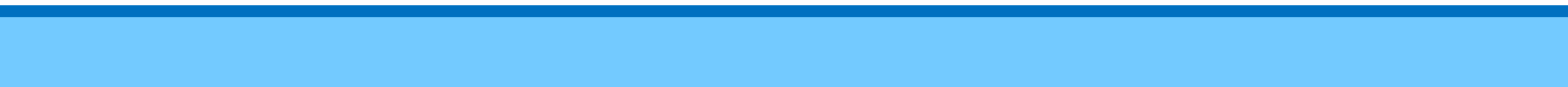
And more...



Recap - what have we learned?



1. An overview of the **structure and architecture** of Flutter applications
2. A **basic understanding** of some of the main widgets.
You can then Google or AI other widgets yourself
3. Some **hands-on experience** in creating Flutter Applications, Widgets and Logic



Let's connect...

duursporten (lopen, zwemmen, fietsen, roeien, schaatsen).

Laatste jaar in combinatie met studie Bewegingswetenschappen, Rijksuniversiteit Groningen (RUG).

[See more education](#) ▾

Featured Skills & Endorsements

[Add a new skill](#) ✎

Teaching · 3 Youri Pattynama and 2 connections have given endorsements for this skill

TypeScript · 3 Youri Pattynama and 2 connections have given endorsements for this skill

Angular · 2 Momen El Kamri and 1 connection have given endorsements for this skill

[See 26 more skills](#) ▾

Recommendations

[Ask to be recommended](#) ✎

● Messaging ✎ ⚙

<https://www.linkedin.com/in/peterkassenaar/>

Thank you

